

MATPLOTLIB LIBRARY USE CASES IN DATA SCIENCE THROUGH VISUALIZATION

AIM

To understand the use cases of the Matplotlib (and Seaborn) library in data science by visualizing the Iris dataset.

PROGRAM

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv('Iris_Dataset.csv')
print(data)
data.info()
print(data.describe())
print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)
plt.title('Count of each variety')
plt.show()

dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat([dummies, data.iloc[:, [0, 1, 2, 3]]], axis=1)
print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)
plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)
plt.show()

sns.pairplot(data.drop(columns='Id'), hue='variety', height=2)
plt.show()

for column in ['PetalLengthCm', 'PetalWidthCm', 'SepalLengthCm', 'SepalWidthCm']:
    g = sns.FacetGrid(data, hue='variety', height=4)
    g.map(sns.histplot, column).add_legend()
    plt.show()
```

OUTPUT

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.4	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
..	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

	variety
0	Iris-setosa
1	Iris-setosa
2	Iris-setosa

```

3      Iris-setosa
4      Iris-setosa
..      ...
145    Iris-virginica
146    Iris-virginica
147    Iris-virginica
148    Iris-virginica
149    Iris-virginica

```

[150 rows x 6 columns]

<class 'pandas.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	Id	150 non-null	int64
1	SepalLengthCm	150 non-null	float64
2	SepalWidthCm	150 non-null	float64
3	PetalLengthCm	150 non-null	float64
4	PetalWidthCm	150 non-null	float64
5	variety	150 non-null	str

dtypes: float64(4), int64(1), str(1)

memory usage: 7.2 KB

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.057333	3.758000	1.199333
std	43.445368	0.828066	0.435866	1.765298	0.762238
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

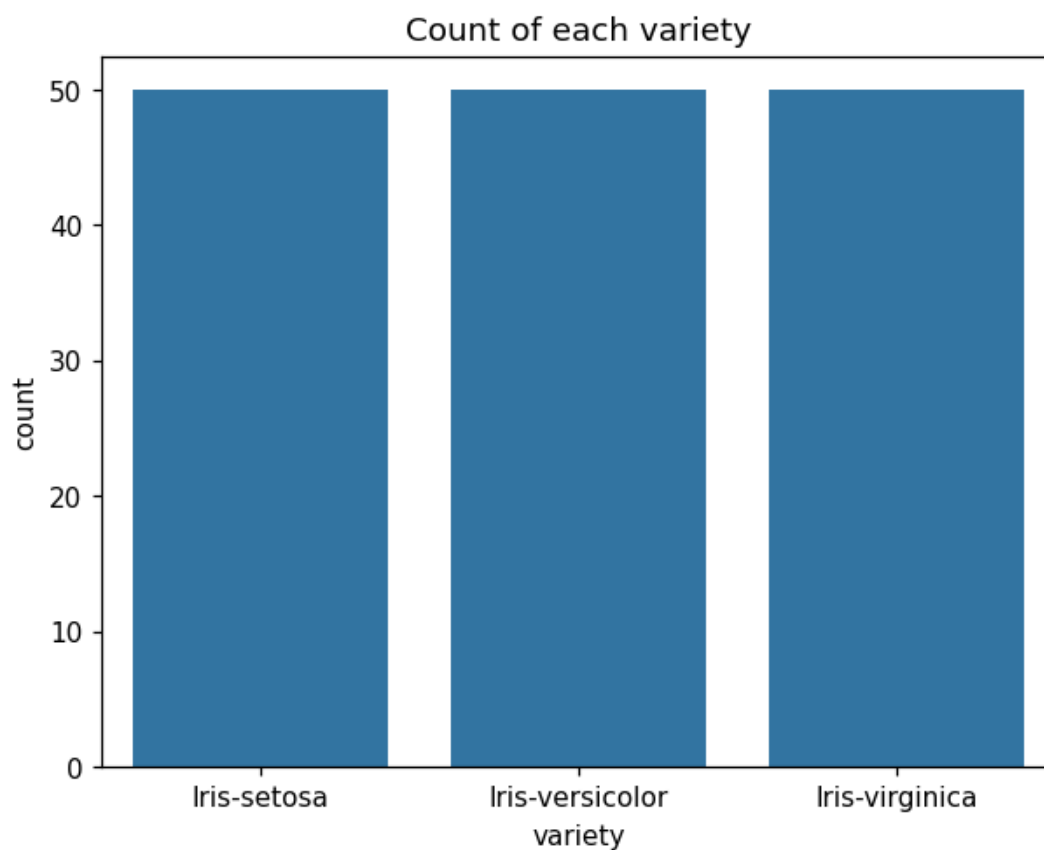
variety

Iris-setosa 50

Iris-versicolor 50

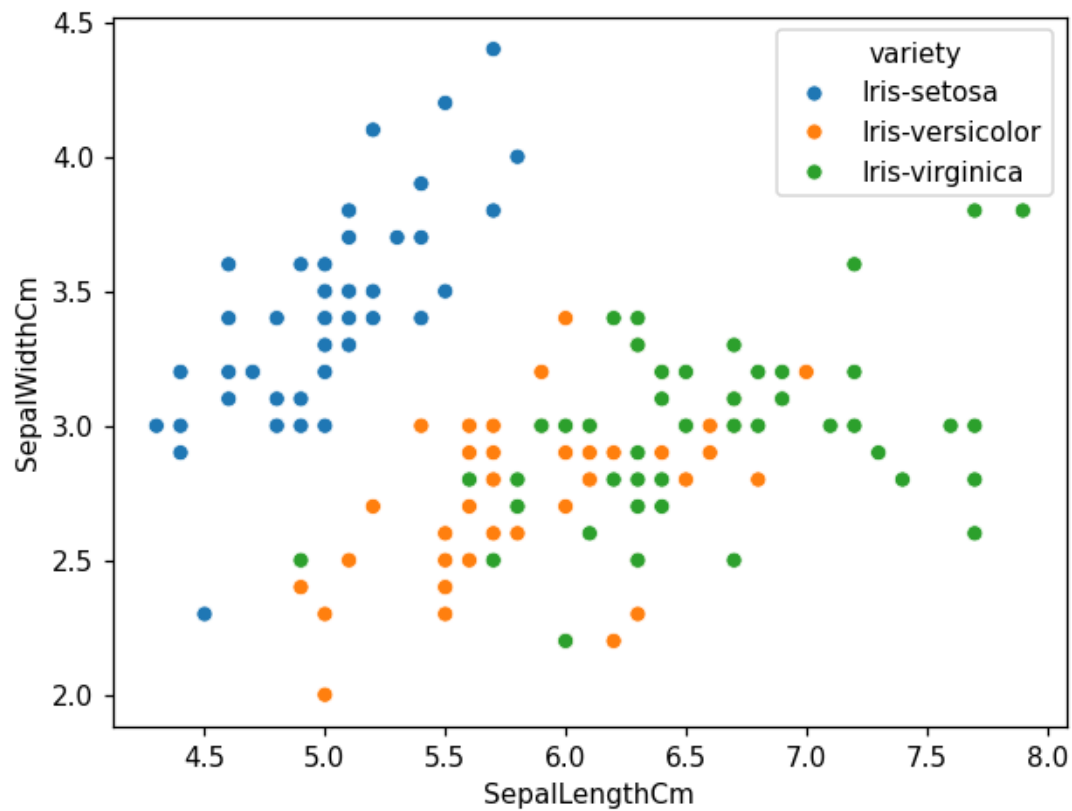
Iris-virginica 50

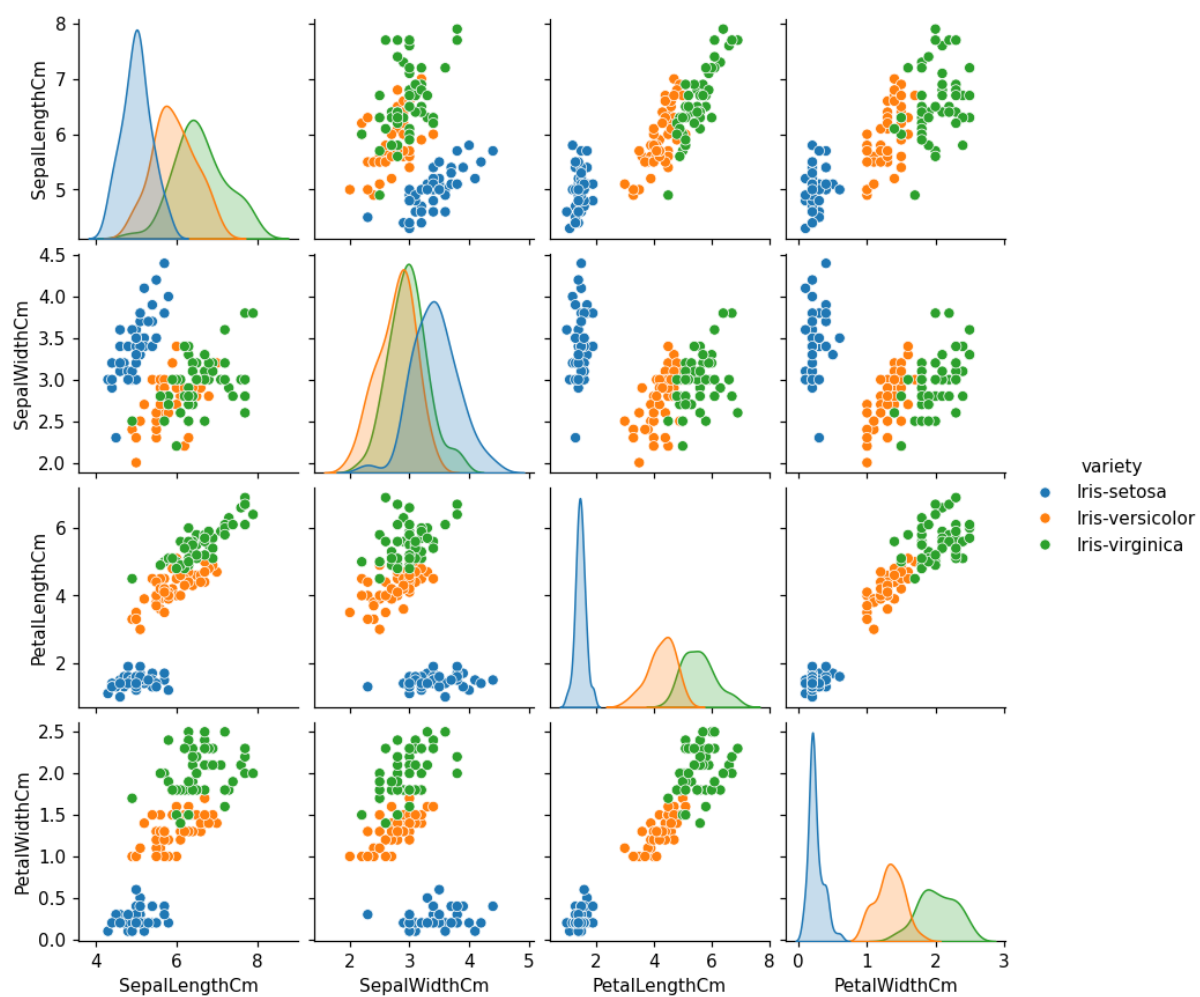
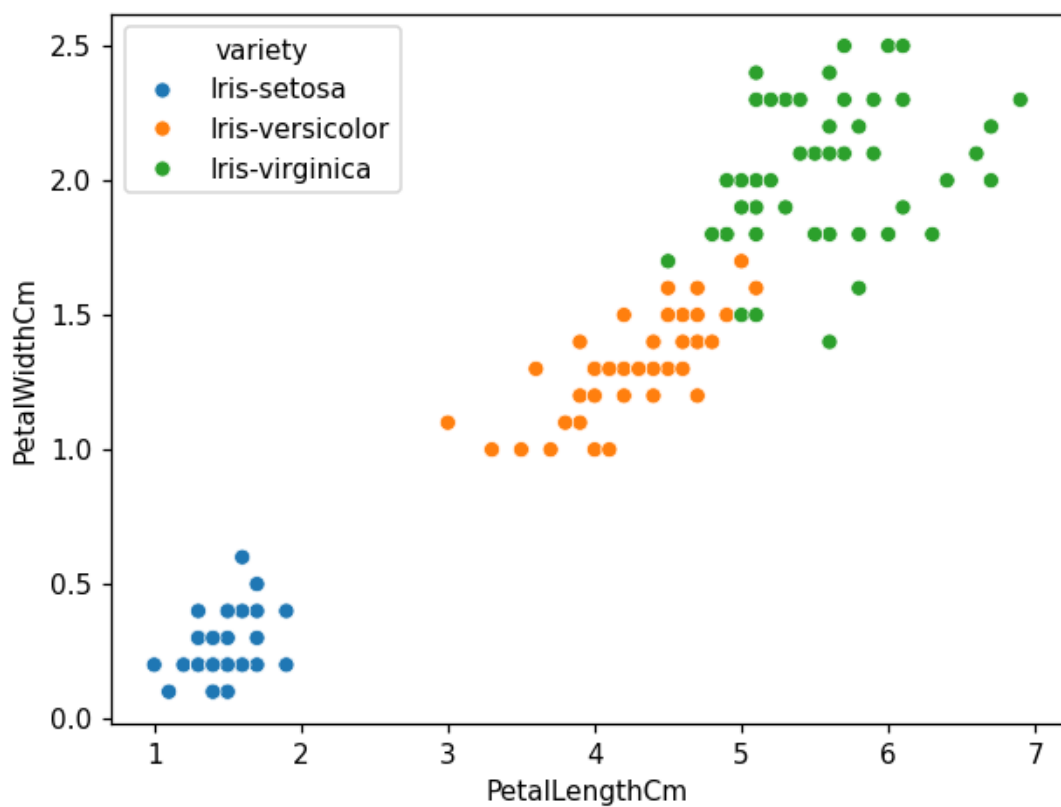
Name: count, dtype: int64

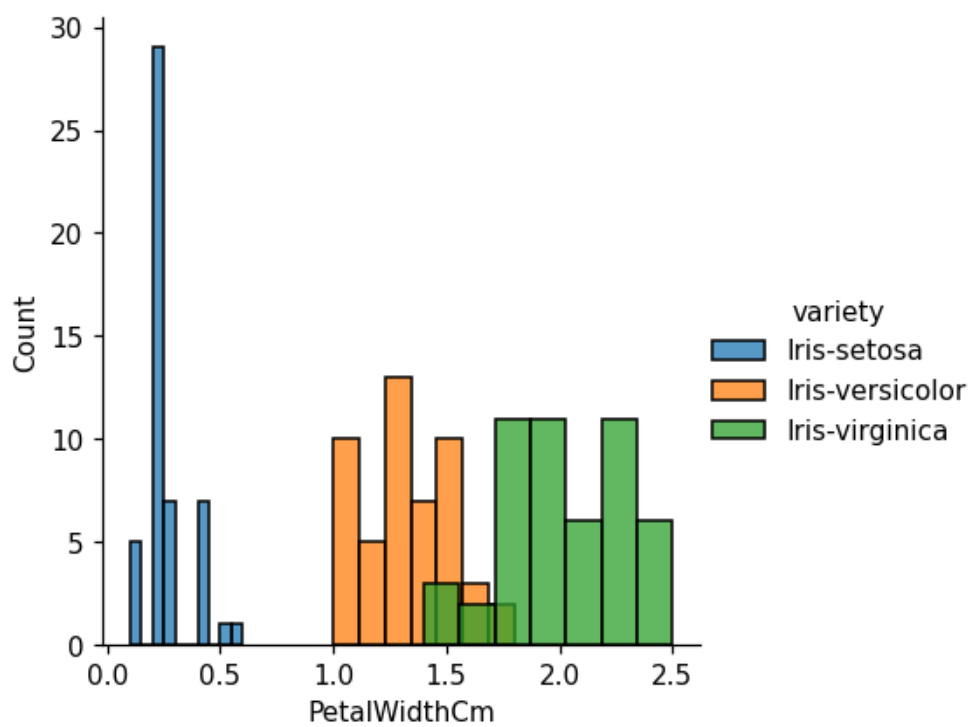
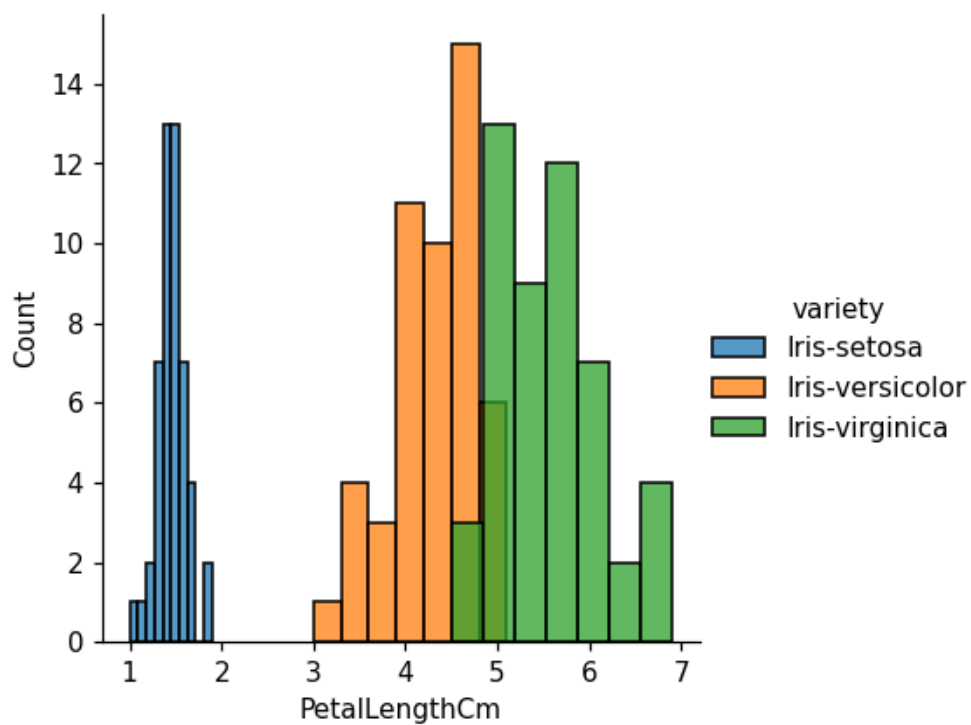


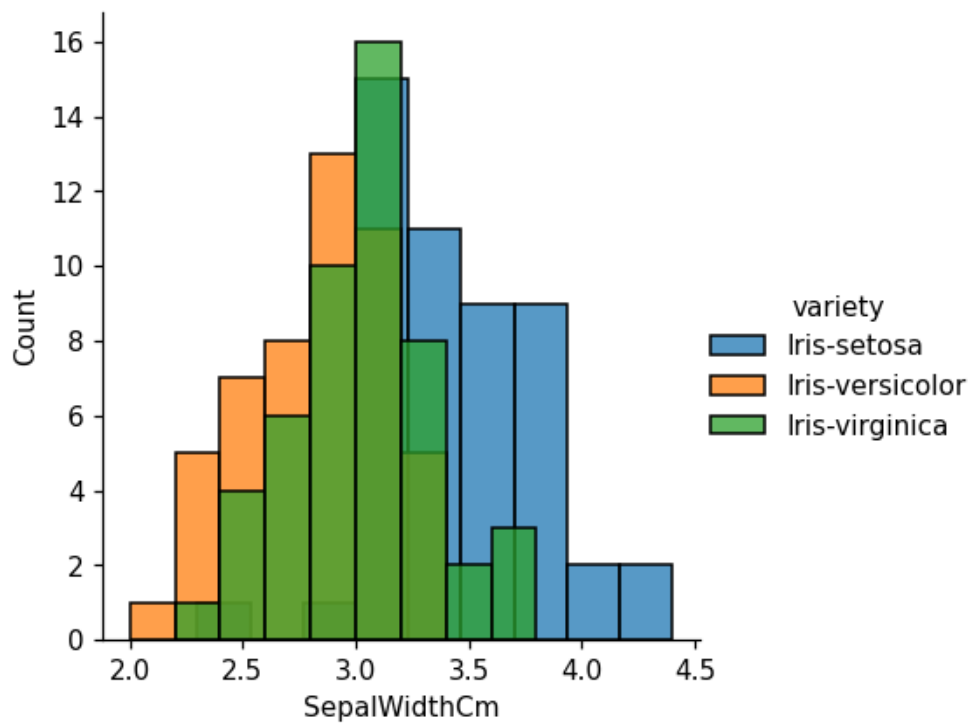
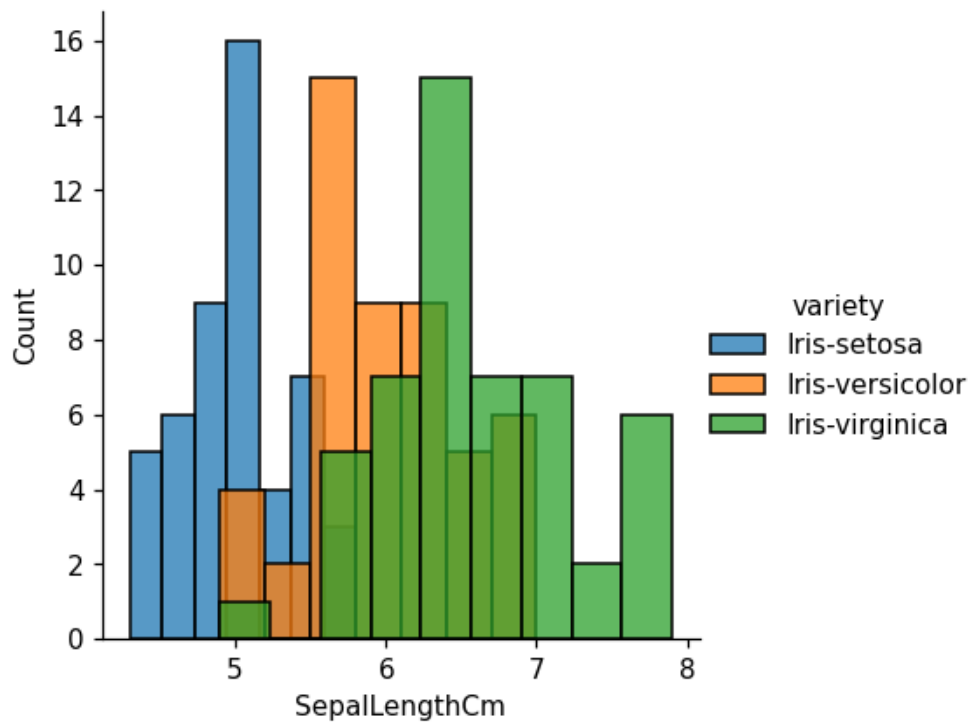
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	\
0	True	False	False	1	5.1	
1	True	False	False	2	4.9	
2	True	False	False	3	4.7	
3	True	False	False	4	4.6	
4	True	False	False	5	5.0	

	SepalWidthCm	PetalLengthCm
0	3.5	1.4
1	3.0	1.4
2	3.2	1.3
3	3.1	1.5
4	3.6	1.4









RESULT

Thus the Iris dataset was explored and visualized with Matplotlib and Seaborn. The plots show that Iris-setosa is clearly separable from the other two varieties using petal length and petal width.